

Contoso Unified Customer Intelligence & Automation Platform

Enterprise Data & Platform Engineering Solution

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1. Problem Framing & Business Objectives

Contoso Services operates across multiple regions and business units using **Dynamics 365 Customer Service**. As the organization scaled, the service platform began facing operational and analytical challenges.

Key Problems Identified

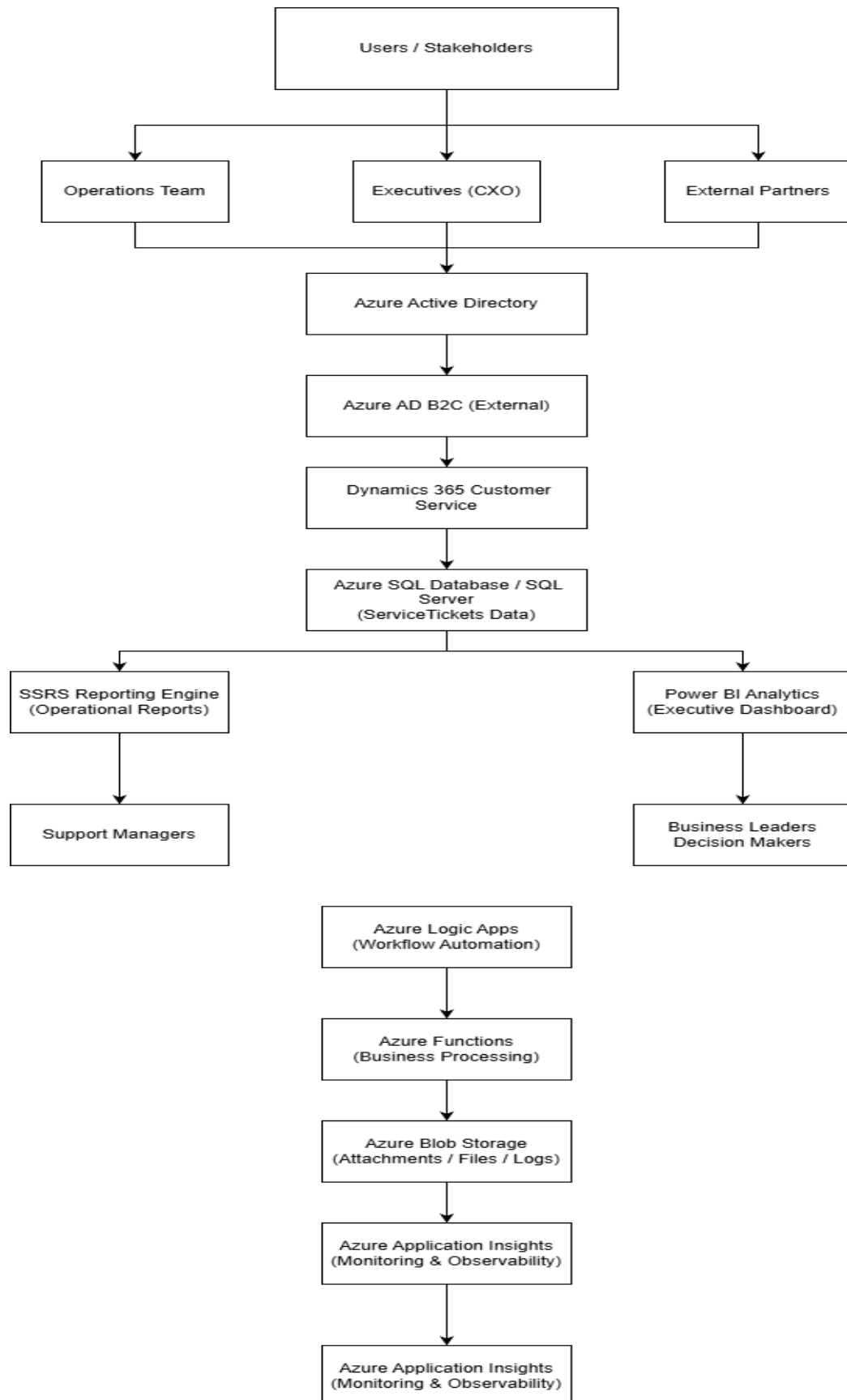
1. Reporting systems are fragmented between operational teams and executives.
2. No standardized KPI definitions exist across regions.
3. Service escalations are handled manually.
4. Attachments stored in CRM increase storage costs.
5. Limited observability of reporting pipelines and automation failures.
6. Security risks when exposing data to external partners.

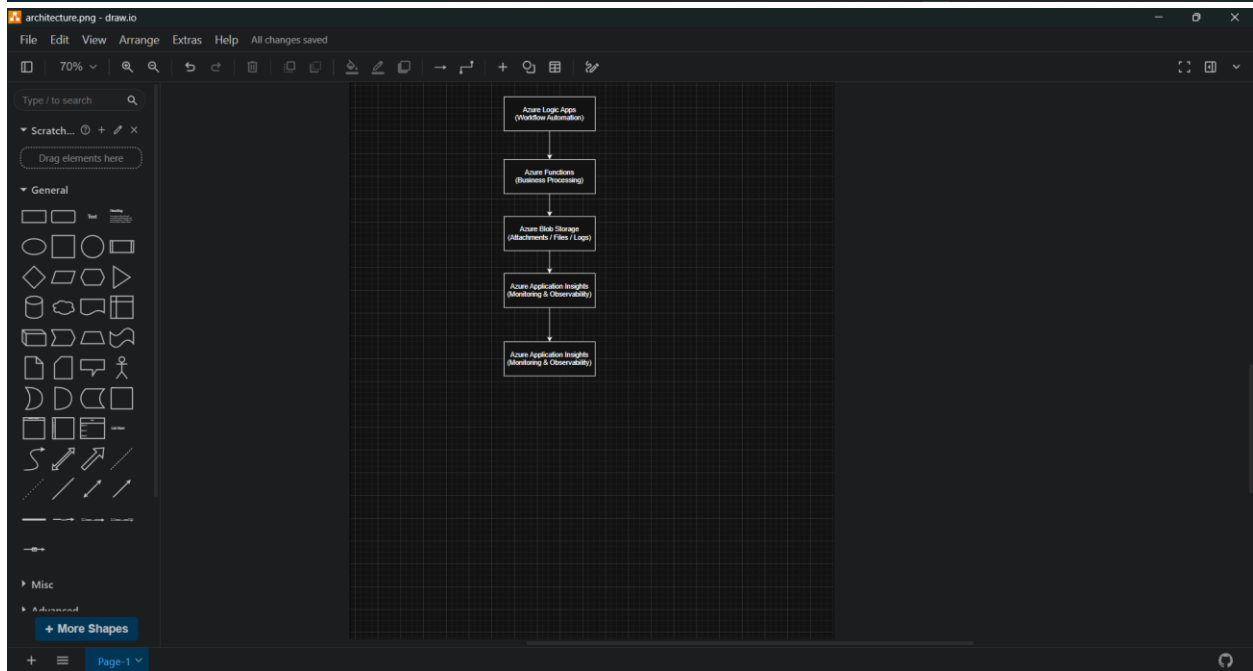
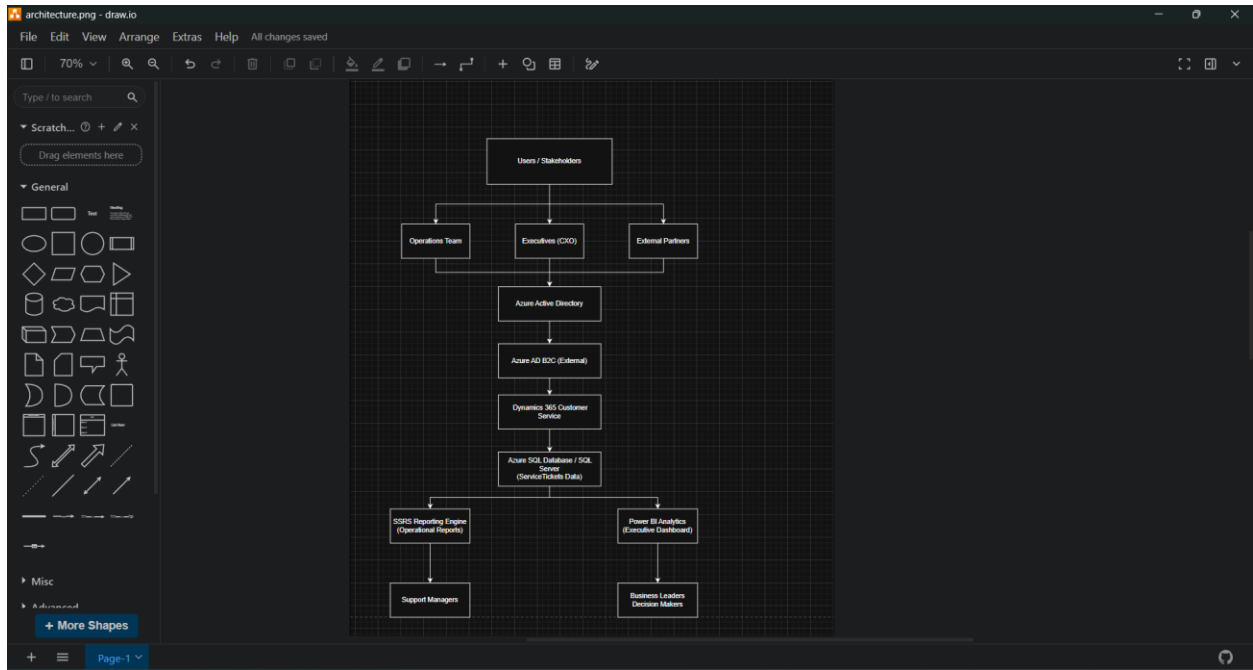
Solution Objective

The objective is to design a **unified service intelligence platform** that:

- Standardizes operational reporting
- Enables executive analytics
- Automates service workflows
- Improves system security
- Introduces monitoring and observability
- Enables data-driven service optimization

The proposed architecture integrates **SSRS, Power BI, Azure services, and Dynamics 365 data sources** into a governed enterprise reporting platform.





2. User Story 1 – Enterprise Reporting Architecture (SSRS)

Objective

Provide a **scalable operational reporting framework** for service teams across regions.

Implementation

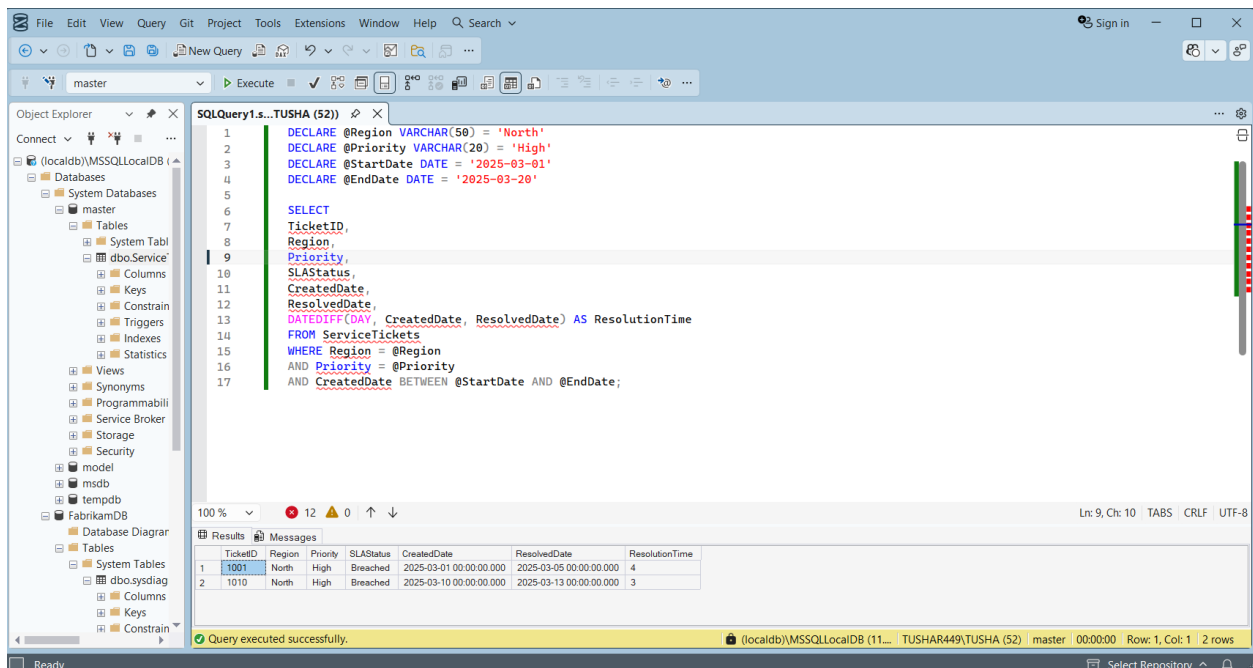
An **SSRS multi-parameter report** was designed using SQL Server Reporting Services.

Parameters Implemented

- Region
- Priority
- SLA Status
- Start Date
- End Date

These parameters allow operational teams to dynamically filter ticket information.

Query Logic Example



The screenshot displays the SQL Server Enterprise Manager interface. The left pane shows the Object Explorer with the 'master' database selected. The central pane shows a SQL query titled 'SQLQuery1.s...TUSHA (52)'. The query defines parameters for Region, Priority, Start Date, and End Date, and then selects ticket information including TicketID, Region, Priority, SLAStatus, CreatedDate, ResolvedDate, and ResolutionTime. The bottom pane shows the results of the query, which are displayed in a table with 2 rows and 7 columns.

```
1 DECLARE @Region VARCHAR(50) = 'North'
2 DECLARE @Priority VARCHAR(20) = 'High'
3 DECLARE @StartDate DATE = '2025-03-01'
4 DECLARE @EndDate DATE = '2025-03-20'
5
6 SELECT
7     TicketID,
8     Region,
9     Priority,
10    SLAStatus,
11    CreatedDate,
12    ResolvedDate,
13    DATEDIFF(DAY, CreatedDate, ResolvedDate) AS ResolutionTime
14 FROM ServiceTickets
15 WHERE Region = @Region
16    AND Priority = @Priority
17    AND CreatedDate BETWEEN @StartDate AND @EndDate;
```

TicketID	Region	Priority	SLAStatus	CreatedDate	ResolvedDate	ResolutionTime
1001	North	High	Breached	2025-03-01 00:00:00.000	2025-03-05 00:00:00.000	4
1010	North	High	Breached	2025-03-10 00:00:00.000	2025-03-13 00:00:00.000	3

Query executed successfully.

Performance Considerations

To improve report performance:

- Indexed columns used: Region, Priority, CreatedDate
- Dataset reused across multiple report components
- Parameter filtering performed at query level to reduce dataset size

Parameter Dependency

Date parameters filter records first, followed by Region and Priority to reduce query load.

Drill-through Capability

Users can click a summary row to open a **detailed ticket report**.

Why SSRS instead of Power BI for this use case

SSRS was selected because:

Reason	Explanation
Operational reporting	Pixel-perfect reports for support teams
Parameterized reports	Easy filtering for operational workflows
Scheduled distribution	Reports can be emailed automatically
High-volume transactional data	SSRS handles structured reporting efficiently

SSRS:

Contoso.rdl [Design] Preview

Region North Priority medium Start Date 2025-03-01 End Date 2025-05-20 View Report

Contoso Service Ticket Report

Region	Ticket ID	Priority	SLAStatus	Created Date	Resolved Date	Resolution Time
North	1006					1
North	1006	Medium	Met	3/6/2025	3/7/2025	1

Output Ready

Contoso.rdl [Design] Preview

Region North Priority high Start Date 2025-03-01 End Date 2025-05-20 View Report

Contoso Service Ticket Report

Region	Ticket ID	Priority	SLAStatus	Created Date	Resolved Date	Resolution Time
North	1001					4
North	1001	High	Breached	3/1/2025	3/5/2025	4
North	1010	High	Breached	3/10/2025	3/13/2025	3

Output Ready

3. User Story 2 – Executive Analytics & KPI Governance (Power BI)

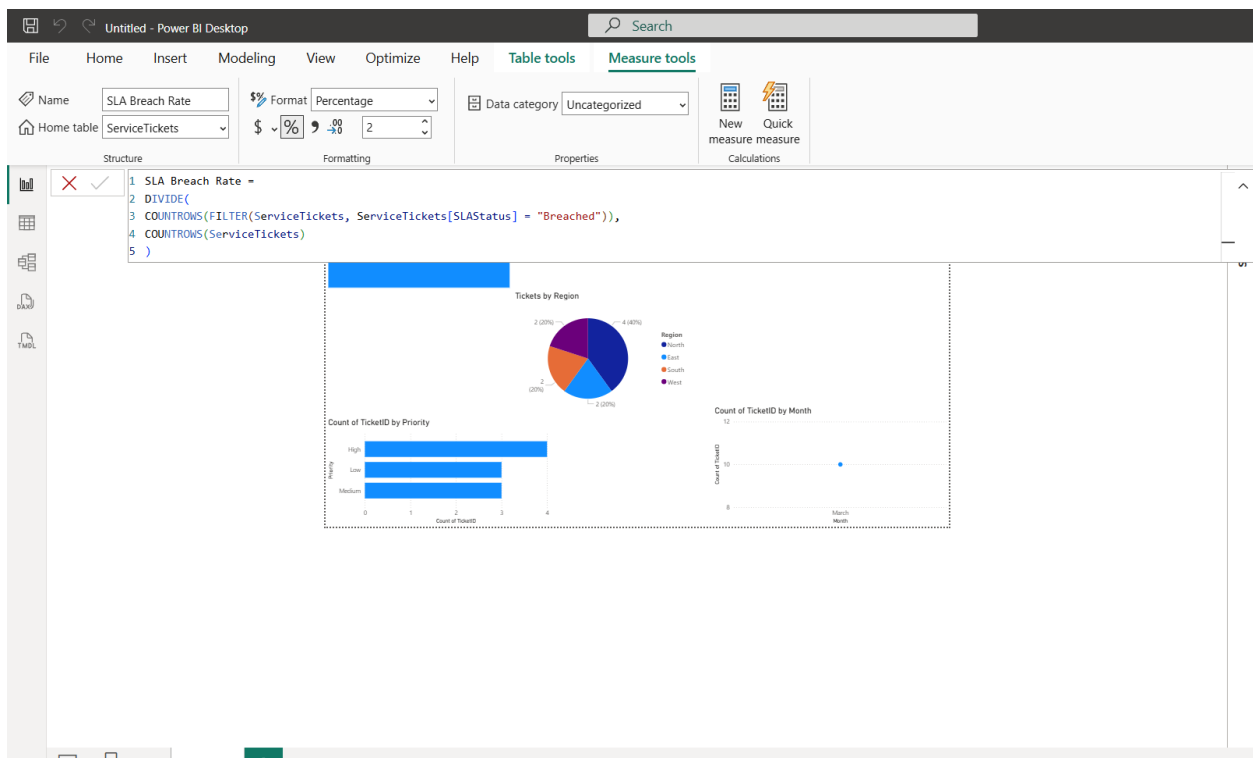
Objective

Enable **data-driven decision making for CXO leadership** using standardized service KPIs.

KPI Definitions

KPI	Definition
Total Tickets	Count of all service tickets
Average Resolution Time	Average number of days to resolve tickets
SLA Breach Rate	Percentage of tickets resolved after SLA
Ticket Distribution by Region	Regional workload analysis
Priority Distribution	Identifies workload severity

Dax Measures:



Dashboard Components

The Power BI dashboard includes:

- KPI cards (Total Tickets, Avg Resolution Time, SLA Breach %)
- Ticket distribution by region
- Ticket trends by month
- Priority distribution

These visualizations help leadership identify service bottlenecks.

Data Storytelling Insights

The dashboard highlights:

- Regions with the highest ticket volume
- Priority levels causing delays
- SLA compliance performance

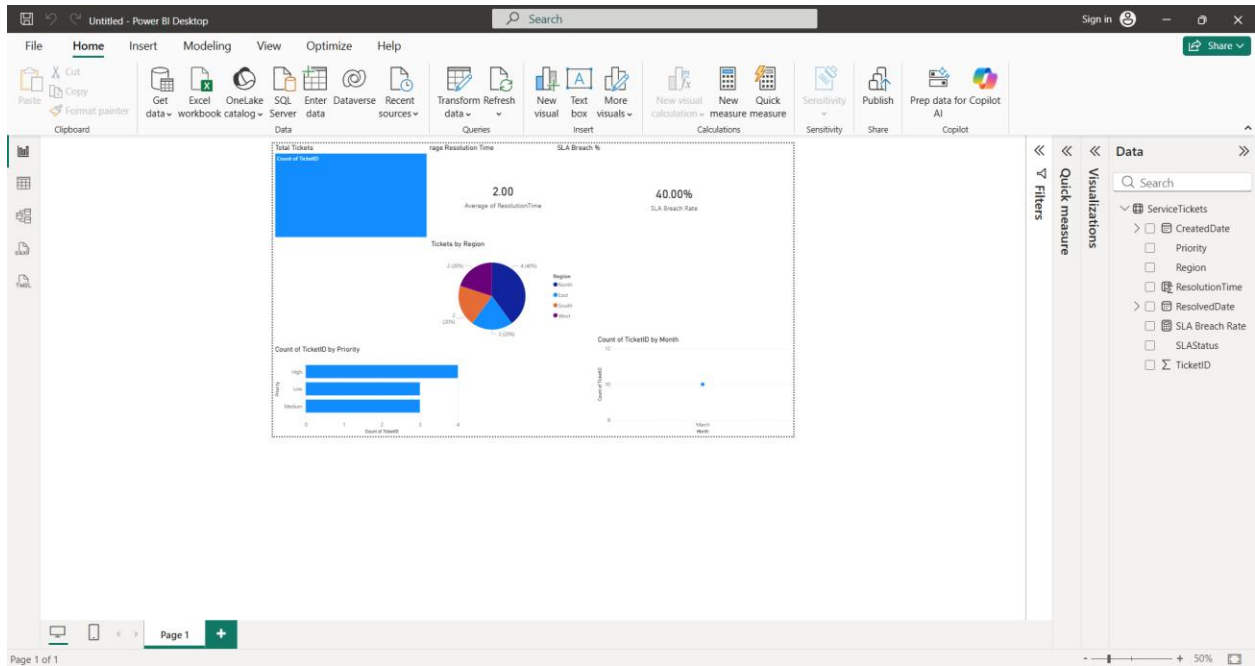
Refresh Strategy

Power BI dataset refresh can be scheduled using:

- Power BI Service scheduled refresh
- Incremental refresh for large datasets

Licensing Considerations

License	Purpose
Power BI Pro	Dashboard sharing
Power BI Premium	Large datasets and enterprise scale



4. User Story 3 – Secure Identity & External Access Strategy

Objective

Provide secure authentication and authorization for both internal employees and external partners.

Identity Architecture

User Type	Identity Provider
Employees	Azure Active Directory
External Partners	Azure AD B2C

Authentication vs Authorization

Authentication verifies **who the user is**, while authorization determines **what resources the user can access**.

Example:

- Azure AD authenticates the user
- Role-based access determines report permissions

Token Based Access

Azure AD issues **JWT access tokens** after successful login.

These tokens allow secure API and service access without exposing credentials.

Role Mapping Strategy

Role	Access
Support Agent	Operational reports
Manager	Region performance
Executive	Power BI dashboards
External Partner	Limited portal access

Security Risks and Mitigations

Risk	Mitigation
Unauthorized access	Role-based access control
Credential theft	Multi-factor authentication
Data exposure	Token-based authentication
External portal risk	Azure AD B2C identity isolation

5. User Story 4 – Advanced Automation & Exception Handling

Objective

Automate service workflows and SLA escalation handling.

Automation Architecture

Automation uses:

- Azure Logic Apps – workflow orchestration
- Azure Functions – business logic execution

SLA Escalation Workflow

1. Ticket created in Dynamics 365
2. Logic App monitors SLA status
3. If SLA breach detected
4. Azure Function processes escalation logic
5. Notification sent to support manager

Failure Handling

Scenario	Handling
API timeout	Retry policy
Function failure	Error logging
Integration failure	Alert triggered

Retry Strategy : Logic Apps retry failed operations with exponential backoff.

Cost and Scaling

Serverless architecture ensures cost efficiency because resources scale automatically based on demand.

When NOT to Use Serverless

Serverless may not be ideal when:

- Long-running workloads exist
- Real-time processing is required
- High compute workloads are continuous

6. User Story 5 – Data Storage, Secrets & Observability

Objective

Implement secure storage, secrets management, and system monitoring.

Azure Blob Storage

Used for storing:

- Ticket attachments
- Logs
- Exported files

Benefits:

- Reduces CRM storage cost
- Scalable storage architecture

Azure Key Vault

Key Vault securely stores:

- API keys
- Database connection strings
- Authentication secrets

Applications retrieve secrets using managed identity.

Azure Application Insights

Provides observability through:

- Application performance monitoring
- Error tracking
- Request telemetry

Example Monitoring Metrics

- API response time
- Failed automation workflows
- Report execution time

Alerting Scenarios

Alerts are triggered when:

- SLA breach rate exceeds threshold
- Automation failures occur
- Application errors increase

Trade-offs

Component	Limitation
Blob Storage	Requires governance policies
Key Vault	Access configuration complexity
Application Insights	Cost increases with large telemetry volume

7. User Story 6 – Analytical Thinking & Service Optimization

Analytical Questions

1. Which region generates the highest ticket volume?
2. What priority tickets cause SLA breaches?
3. How long does it take to resolve tickets on average?

Observed Trends

Analysis of the dataset revealed:

- High priority tickets often contribute to SLA breaches.
- Certain regions generate higher support demand.
- Ticket resolution time varies significantly by priority.

Root Cause Analysis

Potential root causes include:

- Uneven ticket distribution across regions
- Insufficient support staff during peak workloads
- Delays in high priority issue escalation

Business Recommendations

1. Introduce automated SLA escalation workflows.
2. Allocate support resources based on ticket volume trends.
3. Implement KPI monitoring dashboards for leadership.
4. Improve knowledge base to reduce repetitive tickets.

Data Limitations

- Dataset represents a limited timeframe.
- Ticket categories are not included.
- Customer satisfaction data is unavailable.

Future analytics could incorporate:

- Customer sentiment analysis
- Ticket category classification
- Predictive SLA breach detection

8. Risks & Future Improvements

Risks

- Integration complexity between multiple Azure services
- Data governance challenges
- Monitoring cost growth

Future Enhancements

- AI-based ticket classification
- Predictive service analytics
- Automated service capacity planning

9. Conclusion

The proposed **Contoso Unified Customer Intelligence Platform** integrates reporting, analytics, automation, and monitoring into a single governed architecture.

Key outcomes include:

- Standardized enterprise reporting using SSRS
- Executive analytics through Power BI dashboards
- Secure identity and partner access through Azure AD and Azure AD B2C
- Automated service workflows with Logic Apps and Azure Functions
- Centralized storage and observability using Azure Blob Storage and Application Insights
- Actionable insights that support continuous service improvement

This architecture enables Contoso to move from **basic reporting to a decision-driven service intelligence platform**.